Data Analysis Programming
Semester 2026-I
Workshop No. 1 — Project Definition and Source Research

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Welcome to the first workshop of the *Data Analysis Programming* course!

This workshop marks the beginning of a semester-long project around the broad topic of **Public Opinion & Sentiment Analysis**. Each team will define their own specific angle within this topic, identify appropriate data sources, and produce a comprehensive project definition document that will serve as the foundation for all subsequent workshops.

The central idea is to build a complete, end-to-end data engineering and NLP pipeline — from raw data ingestion to an interactive dashboard — that allows a functional user to understand what people think and say about a specific subject of interest.

General Workshop Definition: Source research and project definition involve identifying, evaluating, and validating data sources that will feed a data pipeline, while establishing the conceptual and functional foundations of the project. In this workshop, teams will explore open APIs and scrapable websites as data sources, characterize their data, define the extraction strategy, and formalize the project goals, users, and scope through a structured proposal document.

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Workshop Scope and Objectives

- **Topic Angle Definition:** Choose a specific subject within the broad *Public Opinion & Sentiment Analysis* topic that is meaningful, feasible, and rich in available textual data expressing opinions or sentiment.
- **Source Identification and Characterization:** Identify and evaluate at least two open data sources — one REST API and one web scraping target — that provide textual content relevant to the chosen topic.
- **Extraction Strategy Design:** Define the extraction methodology for each source, including data fields to collect, format of raw data, and ingestion periodicity.
- **Data Sample Validation:** Manually collect a small but representative data sample from each source to verify accessibility, data quality, and relevance before committing to the full pipeline.
- **Project Formalization:** Produce a structured project definition document including goals, functional users, user stories, motivation, and limitations.

Source Research and Characterization Methodology

1. **Topic Angle Selection:**
   - Define the specific subject your team will analyze (e.g., climate change, sports, entertainment, technology, gastronomy, music, political discourse)
   - Justify why this topic is relevant and what kind of opinions or sentiment it generates in the public space
   - Identify the language of the data (English or Spanish) and its implications for NLP processing
   - Establish the time scope of interest: is the topic ongoing, periodic, or event-driven?
2. **API Source Research:**
   - Identify at least one open or free-tier REST API that provides data relevant to your topic (e.g., Reddit API, NewsAPI, Bluesky API, Last.fm API, TMDB API, API-Football, Spoonacular API)
   - Document the API: base URL, authentication method, available endpoints, rate limits, data format (JSON), and fields available
   - Justify why this API is appropriate for your topic and what textual or metadata fields will be used for sentiment analysis
   - Verify API accessibility by successfully performing at least one real request and documenting the response structure

3. **Web Scraping Source Research:**
   - Identify at least one publicly accessible website that contains opinion-rich textual content relevant to your topic (e.g., news sites, forums, review platforms, opinion blogs)
   - Verify the site's `robots.txt` and terms of service to ensure scraping is ethically and legally permissible
   - Document the target pages, HTML structure, and fields to be extracted (titles, body text, dates, authors, ratings, comments)
   - Justify why this source complements the API source and what additional perspective it provides on the topic
4. **Data Sample Collection and Validation:**
   - Manually collect a minimum of 20 records from each source (API and scraping)
   - Present the samples in a structured format (JSON for API, table for scraping) clearly showing all relevant fields
   - Perform a basic quality assessment: identify missing fields, encoding issues, language inconsistencies, or data noise
   - Define preliminary data cleaning needs based on the sample observation
   - Confirm that both sources contain sufficient textual content for sentiment analysis and NLP processing
5. **Extraction Strategy Definition:**
   - Define the ingestion periodicity for each source (e.g., hourly, daily, weekly) based on the nature of the topic and data availability
   - Specify the raw data format for persistence: JSON files with datetime-based nomenclature (e.g., `source_topic_YYYYMMDD_HHMMSS.json`)
   - Define the metadata fields to be persisted alongside text content (timestamps, source identifiers, author, URL, category, etc.)
   - Justify the chosen periodicity in relation to the topic dynamics (e.g., a sports topic may require daily ingestion during season)

Project Definition Document

1. **Project Overview:**
   - *Title and Topic Angle:* Clear and descriptive project title reflecting the specific sentiment analysis angle
   - *Motivation:* Why is this topic relevant? What value does analyzing public opinion on this subject provide?

- *Goal:* A concise statement of what the project aims to achieve from a data engineering and NLP perspective
- *Scope and Limitations:* What is included and explicitly excluded from the project; known data, technical, or time constraints

2. **Functional User Definition:**
   - Identify and describe the primary functional user of the final dashboard (e.g., a brand analyst, journalist, researcher, community manager)
   - Define the user's profile: technical level, information needs, and decision-making context
   - Justify why this user benefits from the sentiment analysis pipeline and dashboard

3. **User Stories:**
   - Write at least five user stories in standard format: *"As a [user], I want to [action] so that [benefit]"*
   - Cover both governance needs (data quality visibility) and storytelling needs (insight discovery)
   - Prioritize user stories by importance to guide dashboard design in later workshops

4. **Pipeline Architecture Overview:**
   - Provide a high-level diagram of the proposed data pipeline: Bronze (raw JSON) → Silver (processed Parquet) → Gold (aggregated summaries)
   - Identify the tools and technologies to be used at each layer (Apache Airflow, Python, Pandas/Polars, PySpark, Plotly Dash, Docker)
   - Describe the orchestration strategy: Airflow DAGs for Bronze ingestion and FileSensor-triggered DAG for Silver processing

Deliverables and Documentation

1. **Project Definition Report (PDF):**
   - Complete document covering all sections described above: topic angle, source characterization, data samples, extraction strategy, project definition, functional user, user stories, and pipeline overview
   - Must include screenshots or JSON snippets of real API responses and scraped content as evidence of source validation
   - All diagrams must be clearly labeled and referenced in the text

2. **Data Samples:**
   - At least 20 raw JSON records from the API source

- At least 20 structured records from the web scraping source (CSV or JSON format)
- Both samples must be included in the GitHub repository

3. **Repository Setup:**

- Create the course GitHub repository with the following base folder structure:
  - `datalake_bronze/` — for raw JSON ingestion outputs
  - `datalake_silver/` — for processed Parquet files
  - `datalake_gold/` — for aggregated summary files
  - `airflow/dags/` — for Airflow DAG definitions
  - `dashboard/` — for Plotly Dash applications
  - `notebooks/` — for exploratory analysis notebooks
  - `workshop_1/` — for this workshop's deliverables and data samples
- Provide a comprehensive `README.md` documenting the project topic, team members, sources, and repository structure

Submission Requirements

- Submit your complete Project Definition Report as a single PDF through the designated course platform
- Include your GitHub repository link with the initial structure and data samples
- All documentation must be in **English** and follow professional technical writing standards
- Ensure proper citations for APIs, websites, and any referenced frameworks or methodologies
- The data sample must demonstrate that both sources are accessible and contain suitable content for NLP and sentiment analysis

Important Guidelines and Considerations

- **Source Validation is Critical:** Do not commit to a source without first verifying it works. A source that seems ideal on paper but is inaccessible in practice will block all subsequent workshops. The data sample is your proof of concept.
- **Open and Free Sources Only:** All APIs and websites must be freely accessible without paid subscriptions. Prefer APIs with free tiers and open registration.
- **Ethical Scraping:** Respect `robots.txt`, avoid overloading servers, and do not scrape content that violates privacy or terms of service. Document your ethical assessment of each scraping source.
- **NLP Feasibility:** Ensure your sources contain enough free text (comments, reviews, articles, posts) to support sentiment analysis. Purely structured or numeric data is not suitable.
- **Topic Originality:** Each team must work on a different topic angle. Coordinate with your professor to confirm your angle is unique within the course group.
- **Foundation for Future Workshops:** This workshop defines the foundation for your entire project. Invest time in selecting good sources and defining a clear, realistic scope — changes at this stage are much less costly than changes in later workshops.

Good luck, and start your data engineering journey by choosing a topic that genuinely interests your team — curiosity about the subject will make every subsequent workshop more meaningful and engaging!